

$P_1 - P_2 - P_3 - P_4 - P_5$

Week 8:-

(1) For the following sequence of pages, using FIFO, LRU, MRU and optimal page replacement algorithms page no is 701203042303032)

No' of frames in physical memory are: 4

Page	r_1	r_2	r_3	H/M
3	3	-	-	m
2	3	2	-	m
1	3	2	1	m
3	3	2	1	H
4	4	2	1	m
1	4	2	1	H
6	4	6	1	m
2	4	6	2	m
4	4	6	2	H
3	3	6	2	m
4	3	4	2	m
2	3	4	2	H
1	3	4	1	m
4	3	4	1	H
5	5	4	1	m
2	5	2	1	m
1	5	2	1	H
3	5	2	3	m
4	4	2	3	m

FIFO :- Hits = 6 , miss = 13

$$\text{miss} = 1 - \frac{m}{7}$$

→ LRU : replaces the page that has been unused for the longest time

Hits = 5 , miss = 14

→ MRU = MRU replaces the page that was used most recently

Hits = 7 , miss = 12

→ Optimal : optimal replaces the page whose next use is farthest in the future

hits = 9, miss = 10

Page replacement algorithm	Hits	misses
FIFO	9	10
MRU	9	11
LRU	9	11
optimal	9	10

Week-8 (3) :- Apply FIFO page replacement algorithm for the following sequence of pages and check whether Belady's algorithm or Belady's anomaly exist or not, also for checking for Belady's anomaly take the physical memory of 3 frames or 4 frames the sequence is 1 2 3 4 1 2 5 1 2 3 4 5.

Week-8 (3) answer :- Given 1 2 3 4 1 2 5 1 2 3 4 5
 → FIFO with 3 frames

Page	Frame 1	Frame 2	Frame 3	H/M
1	1	-	-	H
2	1	2	-	H
3	1	2	3	H
4	4	2	3	M
1	4	1	3	M
2	4	1	3	M
5	5	1	2	M
1	5	1	2	M
2	5	1	2	M
3	5	3	2	M
4	5	3	2	M
5	5	3	2	M
	5	5	2	M

Page faults = 9

hits = 3

→ FIFO with 4 frames.

Page	Frame 1	Frame 2	Frame 3	Frame 4	Hit
1	1	1	1	1	m
2	4	2	1	1	m
3	4	2	3	1	m
4	1	2	3	4	m
1	1	2	3	4	m
2	1	2	3	4	m
5	5	2	3	4	4
1	5	2	3	4	m
2	5	1	3	4	m
3	5	1	3	4	m
4	4	2	2	4	m
5	4	5	2	3	m

Page fault 2 10
 Hit = 2
 10 - 2 = 8
 8 / 4 = 2
 2 * 2 = 4
 4 * 2 = 8
 8 * 2 = 16
 16 * 2 = 32
 32 * 2 = 64
 64 * 2 = 128
 128 * 2 = 256
 256 * 2 = 512
 512 * 2 = 1024
 1024 * 2 = 2048
 2048 * 2 = 4096
 4096 * 2 = 8192
 8192 * 2 = 16384
 16384 * 2 = 32768
 32768 * 2 = 65536
 65536 * 2 = 131072
 131072 * 2 = 262144
 262144 * 2 = 524288
 524288 * 2 = 1048576
 1048576 * 2 = 2097152
 2097152 * 2 = 4194304
 4194304 * 2 = 8388608
 8388608 * 2 = 16777216
 16777216 * 2 = 33554432
 33554432 * 2 = 67108864
 67108864 * 2 = 134217728
 134217728 * 2 = 268435456
 268435456 * 2 = 536870912
 536870912 * 2 = 1073741824
 1073741824 * 2 = 2147483648
 2147483648 * 2 = 4294967296
 4294967296 * 2 = 8589934592
 8589934592 * 2 = 17179869184
 17179869184 * 2 = 34359738368
 34359738368 * 2 = 68719476736
 68719476736 * 2 = 137438953472
 137438953472 * 2 = 274877906944
 274877906944 * 2 = 549755813888
 549755813888 * 2 = 1099511627776
 1099511627776 * 2 = 2199023255552
 2199023255552 * 2 = 4398046511104
 4398046511104 * 2 = 8796093022208
 8796093022208 * 2 = 17592186044416
 17592186044416 * 2 = 35184372088832
 35184372088832 * 2 = 70368744177664
 70368744177664 * 2 = 140737488355328
 140737488355328 * 2 = 281474976710656
 281474976710656 * 2 = 562949953421312
 562949953421312 * 2 = 1125899906842624
 1125899906842624 * 2 = 2251799813685248
 2251799813685248 * 2 = 4503599627370496
 4503599627370496 * 2 = 9007199254740992
 9007199254740992 * 2 = 18014398509481984
 18014398509481984 * 2 = 36028797018963968
 36028797018963968 * 2 = 72057594037927936
 72057594037927936 * 2 = 144115188075855872
 144115188075855872 * 2 = 288230376151711744
 288230376151711744 * 2 = 576460752303423488
 576460752303423488 * 2 = 1152921504606846976
 1152921504606846976 * 2 = 2305843009213693952
 2305843009213693952 * 2 = 4611686018427387904
 4611686018427387904 * 2 = 9223372036854775808
 9223372036854775808 * 2 = 18446744073709551616
 18446744073709551616 * 2 = 36893488147419103232
 36893488147419103232 * 2 = 73786976294838206464
 73786976294838206464 * 2 = 147573952589676412928
 147573952589676412928 * 2 = 295147905179352825856
 295147905179352825856 * 2 = 590295810358705651712
 590295810358705651712 * 2 = 1180591620717411303424
 1180591620717411303424 * 2 = 2361183241434822606848
 2361183241434822606848 * 2 = 4722366482869645213696
 4722366482869645213696 * 2 = 9444732965739290427392
 9444732965739290427392 * 2 = 18889465931478580854784
 18889465931478580854784 * 2 = 37778931862957161709568
 37778931862957161709568 * 2 = 75557863725914323419136
 75557863725914323419136 * 2 = 151115727451828646838272
 151115727451828646838272 * 2 = 302231454903657293676544
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